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Completed	Saturday, 17 October 2026, 11:16 AM
Duration	6 mins 52 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Correct

Mark 1.00 out of 1.00

_____sort is the simplest sorting algorithm that works by repeatedly swapping the adjacent elements in case they are unordered in n-1 passes.

a. Bubble ✓

b. Selection

c. Complexity

d. Insertion

Question **2**

Correct

Mark 1.00 out of 1.00

What happens when the element is found in linear search?

a. The search continues until the end of the list

b. The search stops immediately ✓

c. The search backtracks to find duplicate elements

d. The search starts over from the beginning

Question **3**

Correct

Mark 1.00 out of 1.00

Which of the following is not a limitation of binary search algorithm?

- a. Binary search algorithm is not efficient when the data elements more than 1500 ✓
- b. Requirement of sorted array is expensive when a lot of insertion and deletions are needed
- c. Must use a sorted array
- d. There must be a mechanism to access middle element directly

Question **4**

Correct

Mark 1.00 out of 1.00

If a list contains 1000 elements, how many comparisons would a binary search typically make in the worst case?

- a. 500
- b. 10 ✓
- c. 100
- d. 1000

Question **5**

Correct

Mark 1.00 out of 1.00

In binary search, if the target element is less than the middle element, where does the search continue?

- a. At the beginning of the list
- b. In the left sublist ✓
- c. In the right sublist
- d. In the entire list

Question **6**

Correct

Mark 1.00 out of 1.00

What is the best-case time complexity of linear search?

- a. O👎
- b. O(log n)
- c. O(n log n)
- d. O(1) ✓

Question **7**

Correct

Mark 1.00 out of 1.00

Which of the following is a type of searching method?

- a. Merge search
- b. Bubble search
- c. Quick search
- d. Linear search ✓

Question **8**

Correct

Mark 1.00 out of 1.00

In which situation is linear search more efficient than binary search?

- a. When the list is small and sorted
- b. When the list is large and unsorted
- c. When the list is large and sorted
- d. When the list is small and unsorted ✓

Question **9**

Correct

Mark 1.00 out of 1.00

Given an array arr = {45,77,89,90,94,99,100} and key = 100; What are the mid values(corresponding array elements) generated in the first and second iterations?

- a. 90 and 100
- b. 89 and 94
- c. 94 and 99
- d. 90 and 99 ✓

Question **10**

Correct

Mark 1.00 out of 1.00

During a linear search, what is the maximum number of comparisons needed to find an element in a list of size n?

a. n-1
b. n ✓
c. log n
d. n/2

Question **11**

Correct

Mark 1.00 out of 1.00

The average case occurs in the linear search algorithm

a. When the item is not the array at all
b. Item is the last element in the array or item is not there at all
c. When the item is somewhere in the middle of the array ✓
d. When the item is the last element in the array

Question **12**

Correct

Mark 1.00 out of 1.00

What is the key characteristic of binary search?

a. It works on unsorted lists
b. It can be applied only if the list is sorted ✓
c. It compares elements sequentially
d. It always starts from the beginning of the list

Question **13**

Correct

Mark 1.00 out of 1.00

What is the time complexity of binary search in the worst case?

a. O👎
b. O(n log n)
c. O(1)
d. O(log n) ✓

Question **14**

Correct

Mark 1.00 out of 1.00

In binary search, how is the middle element determined?

- a. By starting from the first element
- b. By using a hash function
- c. By dividing the list length by two ✓
- d. By comparing each element sequentially

Question **15**

Correct

Mark 1.00 out of 1.00

What is the advantage of binary search over linear search?

- a. Binary search has a lower time complexity on large, sorted lists ✓
- b. Binary search can find multiple instances of the target element
- c. Binary search does not require dividing the list
- d. Binary search works on unsorted lists